

Corridors of Opportunity: Corridor-Based Allocation for California Climate Investments

Purpose

This memo describes the analytical workflow used to produce my corridor-based alternative to SB 535's census-tract DAC designation for California Climate Investments (CCI) transportation funding. The goal is to identify Los Angeles County corridors that disproportionately serve residents of disadvantaged communities (DACs), providing a more spatially coherent and equity-driven basis for infrastructure investment prioritization.

Background

SB 535 directs a minimum share of Greenhouse Gas Reduction Fund dollars to “disadvantaged communities” as defined by CalEPA using CalEnviroScreen 4.0, census tracts at or above the 75th percentile statewide on a composite pollution-burden and socioeconomic score. In practice, DAC status has become a de facto eligibility threshold for state transportation funds. Critics note that because of spatial mismatch, many DAC residents must travel beyond their home tracts daily to reach jobs, groceries, and services. **Restricting investment eligibility to home tracts therefore risks funding projects that do not improve the trips that matter most to disadvantaged travelers.**

Data Sources

- Origin-destination data: Longitudinal Employer-Household Dynamics (LEHD) data, constrained to Los Angeles County
- Road network: OpenStreetMap via OSMnx, LA County edges (.gpkg) with link attributes
- DAC designation: CalEnviroScreen 4.0 (OEHHA), LA County tracts at or above 75th percentile statewide
- Zone-to-tract crosswalk: zone_tract_mapping.csv linking TAZ zone IDs to 6-digit census tract codes

Analytical Workflow

Step 1: Origin-Decomposed Traffic Assignment

Following Xu et al. (2024), a per-origin All-or-Nothing (AON) traffic assignment was run in AequilibraE for 923 origin zones representing the top 50% of LA County trip volume by origin. For each origin zone, vehicle trips from the LODES OD table were loaded onto the network and routed via shortest path. Results were saved with link IDs preserved, yielding a table of Passenger Car Equivalents (PCE) by link and origin zone.

Step 2: DAC Origin Identification

Origin zones were crosswalked to Census tracts via zone_tract_mapping.csv. Tracts were matched to CalEnviroScreen 4.0 scores and filtered to LA County tracts at or above the 75th CES percentile statewide, consistent with CalEPA's SB 535 DAC definition. This process yielded 933 DAC origin zones out of 2,495 total, generating 29% of countywide PCE, a plausible result given the spatial concentration of trip generation in high-density, lower-income areas. In other words, just as residential neighborhoods in Los Angeles County are segregated by disadvantage, so are commuter trips.

Step 3: Link-Level DAC PCE Aggregation

For each link in the network, two metrics were computed:

- (1) **total DAC-origin PCE:** the sum of PCE on that link attributable to trips originating in DAC zones
- (2) **DAC PCE share:** the proportion of a link's total traffic volume originating from DAC zones.

These metrics capture both the absolute volume of disadvantaged travelers served and the relative dependence of a link on DAC-origin trips.

Step 4: Composite Priority Score

To rank links and corridors, a composite score was constructed using the same percentile-averaging methodology employed by OEHHA in CalEnviroScreen 4.0. Each link was assigned a percentile rank (0–100) on both DAC PCE and DAC PCE share. The composite score is the unweighted average of the two percentile ranks. Links scoring at or above the 75th percentile on the composite score are designated high-DAC-burden corridors, mirroring the SB 535 threshold logic. These corridors are visualized on the map accessible via <https://alexploratory.github.io/corridorsofopportunity.github.io/>.

Step 5: Corridor Aggregation

Links were joined to OpenStreetMap edge attributes (street name, highway type) and aggregated by named corridor. Connector ramps (motorway_link, primary_link, etc.) were excluded. The composite score was re-percentiled at the corridor level, combining DAC PCE and DAC PCE

share across all constituent links. The resulting corridor ranking is the primary output of the analysis.

My Results

- 933 of 2,495 zones identified as DAC origins (37.4%)
- DAC origins generate 29% of countywide PCE across all modeled origins
- Top-ranked corridors include MLK Jr. Boulevard, Long Beach Boulevard, Rosecrans Avenue, Century Boulevard, and Artesia Boulevard. These are all major arterials in South and Southeast LA, the most densely-populated and largest contiguous DAC cluster in the state of California.
- Top corridors carry DAC-origin PCE shares of 90-95%, compared to the 29% countywide average. This finding confirms a strong geographic concentration between residential disadvantage and road usage by disadvantaged travelers.

Methodological Justifications

Why Average the Percentiles?

Percentile averaging is the same approach OEHHHA uses to combine indicators within CalEnviroScreen components. Using it here creates an explicit methodological parallel to the screening tool this analysis is designed to complement and extend. It also avoids the need to set arbitrary weights between the two metrics, treating volume and share as equally important dimensions of DAC corridor burden.

Why a 75th Percentile Threshold?

The 75th percentile is the SB 535 DAC threshold. Applying the same cutoff to corridors ensures consistency with existing policy and makes the corridor map directly comparable to the Census tract-level DAC map as an alternative eligibility framework.

Why 50% Trip Coverage?

Running a full origin-decomposed assignment for all 2,495 zones would have been computationally intensive (approximately 14 hours for 923 origins). The top 50% of origins by trip volume covers the majority of network flows and is sufficient to identify high-burden corridors, though I'd like to run the analysis again at 100%. Low-volume origins contribute marginally to link-level aggregates and are unlikely to change corridor rankings materially.

Why AON Rather than Equilibrium Assignment?

All-or-Nothing assignment is used because the goal is to trace trip origins to links, not to model congestion effects. Equilibrium assignment distributes flows across alternative routes in response to congestion, which would obscure the origin-to-link attribution that this analysis requires. AON provides a clean, reproducible mapping from each origin zone to the links its trips traverse.

Limitations

First, the travel network underlying this analysis is constrained to the automobile street grid, built from OpenStreetMap road attributes and calibrated to vehicular capacity and speed parameters. Trips made via rail transit, walking, or cycling are not represented. This is a meaningful gap given that SB 535 and related CCI programs frequently prioritize investments in non-single-occupancy vehicle modes, the very modes most used by DAC residents. A multimodal network extension would better capture the corridors where transit-dependent travelers are most concentrated.

Second, the origin-destination inputs are derived from LODES employment data, which captures only home-to-work trips. On average, employment-related travel accounts for less than 20% of all trips an American makes (McGuckin et al). Non-work trips, grocery shopping, medical appointments, school, and recreation, are precisely the trip purposes most relevant to the spatial mismatch argument motivating this analysis. Future iterations of this model should incorporate full-purpose OD data.

Third, the composite DAC priority score reflects both the absolute volume of DAC-origin trips and the share of a corridor's traffic attributable to DAC origins. Major thoroughfares in Downtown Los Angeles, for instance, carry substantial DAC-origin traffic in absolute terms but fall outside the top quartile because they also serve large volumes of higher-income CBD workers, diluting their DAC share. Whether to treat this as a limitation or a feature depends on the policy question at hand, and this parameter sensitivity warrants further exploration.

McGuckin, Nancy, and Nandu Srinivasan. "The Journey-to-Work in the Context of Daily Travel." *Census Transportation Planning Products Program (CTPP)*. American Association of State Highway and Transportation Officials.

Xu, X., Zheng, Z., Hu, Z. et al. A unified dataset for the city-scale traffic assignment model in 20 U.S. cities. *Sci Data* 11, 325 (2024). <https://doi.org/10.1038/s41597-024-03149-8>

Office of Environmental Health Hazard Assessment. 2023. "CalEnviroScreen 4.0."